Exposure Correction

Group 16
Shutter Buddy

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Outline

- Introduction
- Related work
- Dataset
- Platform
- Baseline
- Main Approach
- Evaluation Metric
- Results & Analysis

Introduction

Exposure Correction is used to adjust the brightness of photos or images to achieve an ideal exposure level, correcting detail loss caused by overexposure (too bright) or underexposure (too dark), thereby enhancing image quality and visual effects.

Our work combines the methods from three papers, modifying the network framework to achieve effective exposure correction.

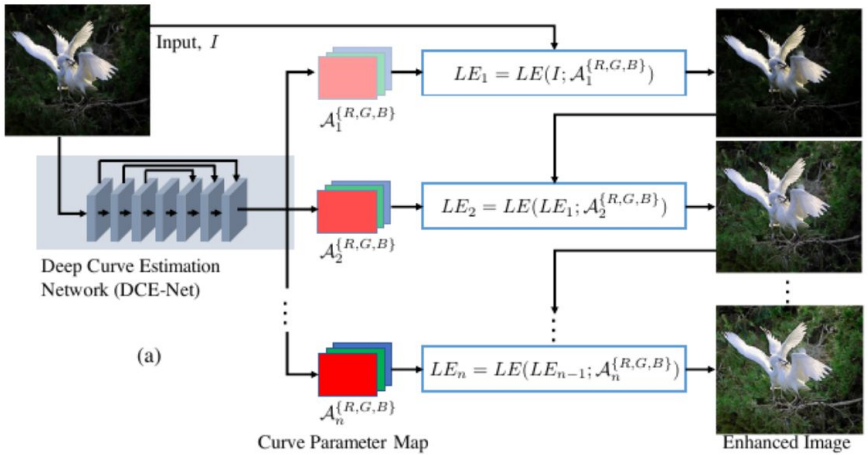
Related work - Zero DCE

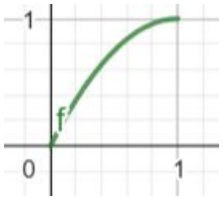
- Light enhancement (effective only for underexposed images)
- Method that can train without ground truth
- lightweight deep network: DCE-net

  input: image

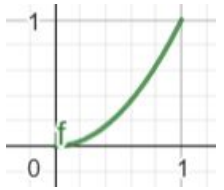
  output: HxWx3x8 element, -1 ~ 1
- similar to gamma correction

LE-curves: $LE(I(x);\alpha) = I(x) + \alpha I(x)(1 - I(x))$



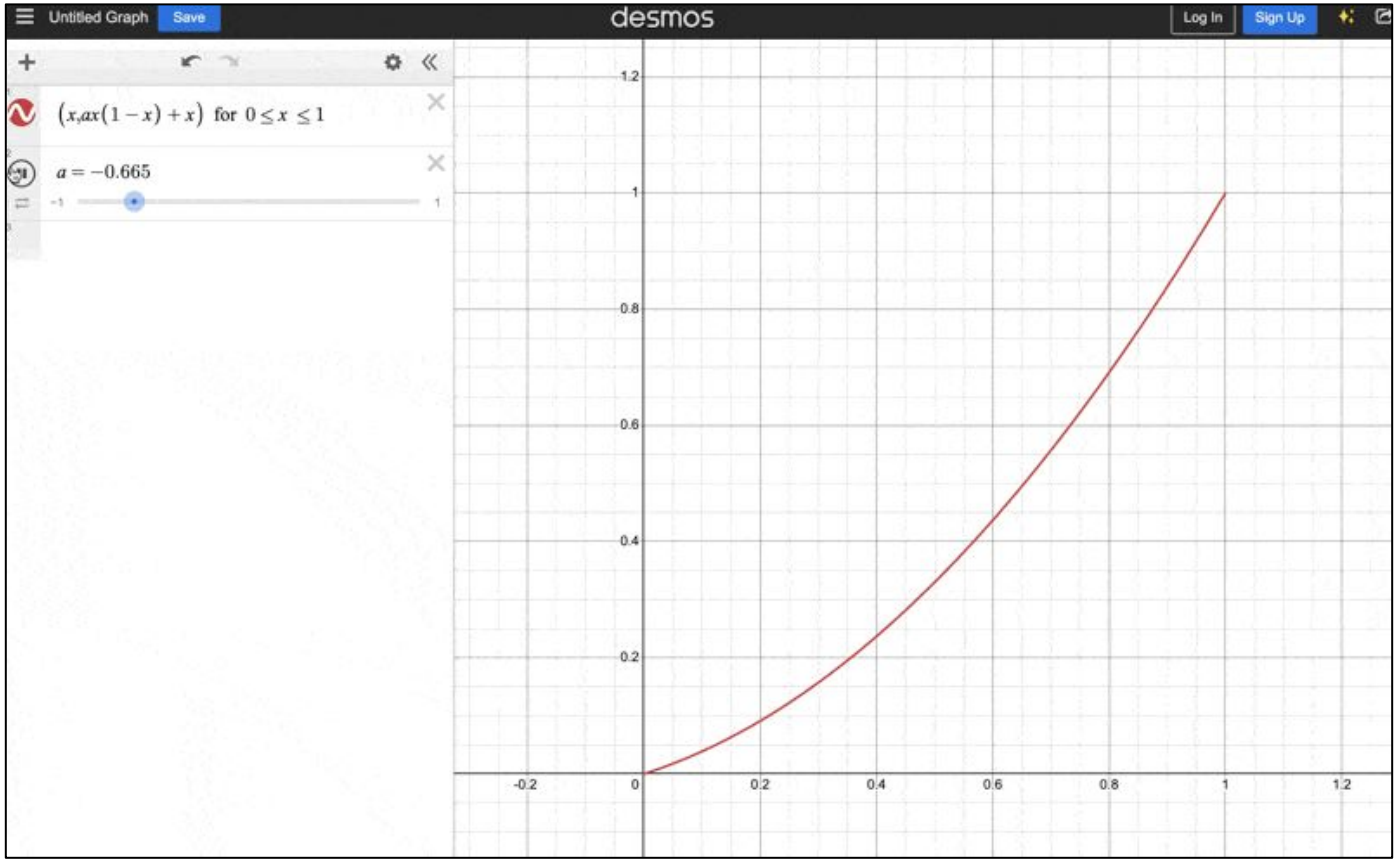


a = 1, brighten



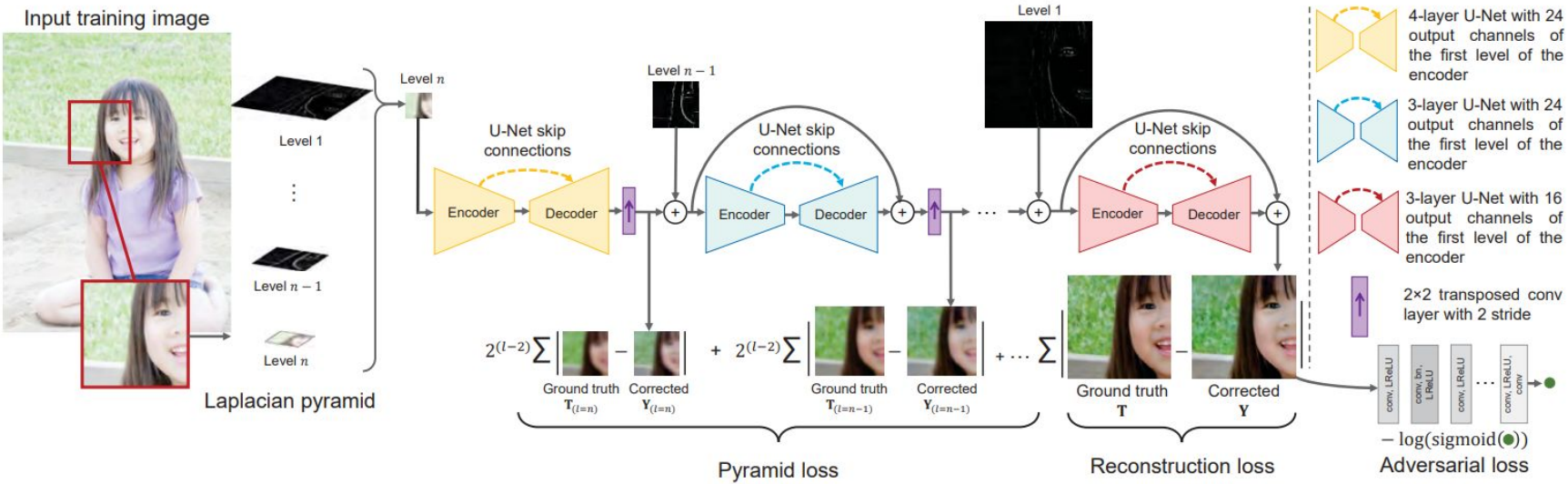
a = -1, darken

LE-curves: $LE(I(x);\alpha) = I(x) + \alpha I(x)(1 - I(x))$



Related work
Learning Multi-Scale Photo Exposure Correction

we reference the framework from this paper



Related work
Reversed and Fused Zero-DCE

Original LE curve from Zero-DCE:

$$LE(x) = I(x) + A(x)I(x)(1 - I(x))$$

Substituting $1 - I_x$ and $1 - O_x$ for I_x and O_x

$$LE(x) = I(x) - A(x)I(x)(1 - I(x))$$

Dataset

- **Source**: Training dataset of Zero_DCE.
- **Composition**:
  - 360 multi-exposure sequences.
  - Total: 2,022 images.
  - All images resized to **512 × 512** resolution.

Zero_DCE training dataset
https://drive.google.com/file/d/1GAB3uGsmAyLgtDBDONbil08vVu5wJcG3/view

Test_data (MIT-Adobe FiveK)
https://paperswithcode.com/dataset/mit-adobe-fivek
https://drive.google.com/drive/folders/1rZGvI3Jd67JUfhaAs9lnDxDfOsLBgemx?usp=drive_link

Platform

Environment

- python = 3.7
- pytorch = 1.13.1
- torchvision = 0.14.1
- torchaudio = 0.13.1
- Cuda = 11.7
- OpenCV

Device

- NVIDIA GeForce RTX 4090
- Ubuntu 24.04 LTS

Baseline

- **What are the chosen baselines?**

  Gamma correction

  Histogram equalization

  Zero DCE

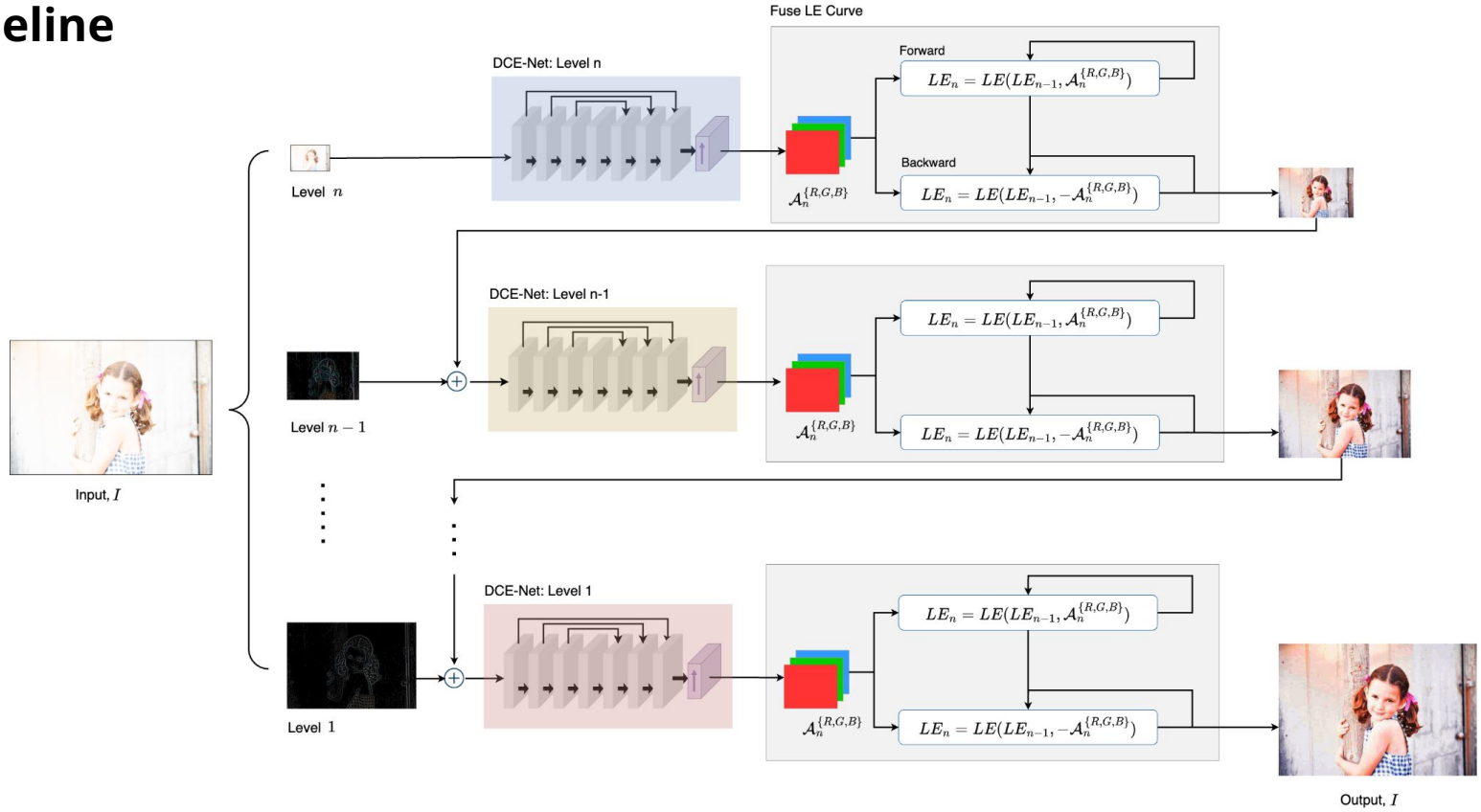
- **How the baseline is implemented**

  Gamma correction, Histogram equalization: codes from hw2

  Zero DCE: codes from GitHub, it can be used directly

Main Approach

Pipeline



Evaluation Metric

- **Qualitative**

  We selected some images with poor exposure for evaluation.

  Using our eyes, we compared and judged which images looked better.

- **Quantitative**

  We use **PSNR** and **SSIM** to measure the similarity between the enhanced image and the ground truth.

  PSNR(↑): Compare the numerical accuracy of each pixel.

  The larger the value, the closer the match.

  SSIM(↑): Based on the human visual system, comparing visual consistency.

  The larger the value, the closer the match.

Results & Analysis - Qualitative

overexpose condition

original



histogram



laplacian_dce



multi laplacian_dce



zero dce



gamma



with fuse



with fuse



Results & Analysis - Qualitative

overexpose condition

original



histogram



laplacian_dce



multi laplacian_dce



zero dce



gamma



with fuse



with fuse



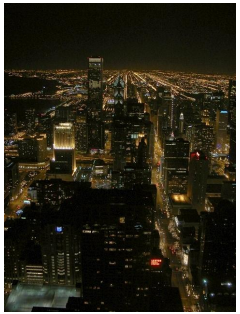
Results & Analysis - Qualitative

underexpose condition

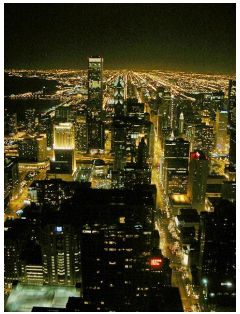
original



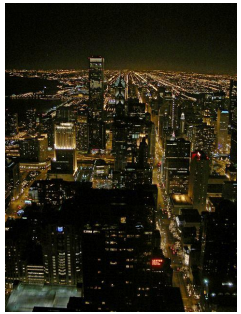
histogram



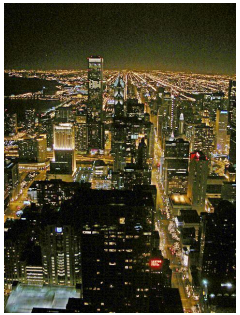
laplacian_dce (backward)



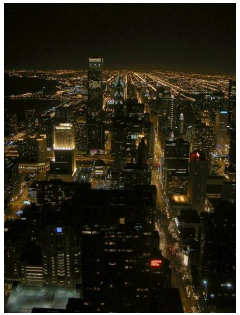
multi laplacian_dce



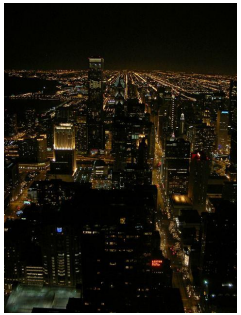
zero dce



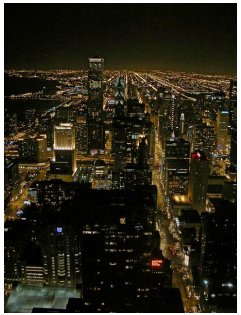
gamma



with fuse



with fuse



Results & Analysis - Qualitative

underexpose condition

original



histogram



laplacian_dce (backward)



multi laplacian_dce



zero dce



gamma



with fuse



with fuse



Results & Analysis - Quantitative underexposed image

PSNR_laplacian_dce	12.22008
PSNR_laplacian_dce_fuse	**17.35632**
PSNR_multi_laplacian_dce	17.27604
PSNR_multi_laplacian_dce_fuse	16.67159
PSNR_histogram	16.90371
PSNR_zero_dce	15.08217
PSNR_laplacian_dce_backward	12.03807

SSIM_laplacian_dce	0.537382
SSIM_laplacian_dce_fuse	**0.867567**
SSIM_multi_laplacian_dce	0.831691
SSIM_multi_laplacian_dce_fuse	0.758748
SSIM_histogram	0.725812
SSIM_zero_dce	0.747626
SSIM_laplacian_dce_backward	0.706475

link to .csv

Results & Analysis - Quantitative overexposed image

PSNR_laplacian_dce	**22.84264**
PSNR_laplacian_dce_fuse	15.8773
PSNR_multi_laplacian_dce	16.75109
PSNR_multi_laplacian_dce_fuse	15.53623
PSNR_histogram	16.86273
PSNR_zero_dce	9.35885
PSNR_laplacian_dce_backward	6.980982

SSIM_laplacian_dce	**0.912495**
SSIM_laplacian_dce_fuse	0.854002
SSIM_multi_laplacian_dce	0.838874
SSIM_multi_laplacian_dce_fuse	0.75821
SSIM_histogram	0.719798
SSIM_zero_dce	0.720766
SSIM_laplacian_dce_backward	0.601872

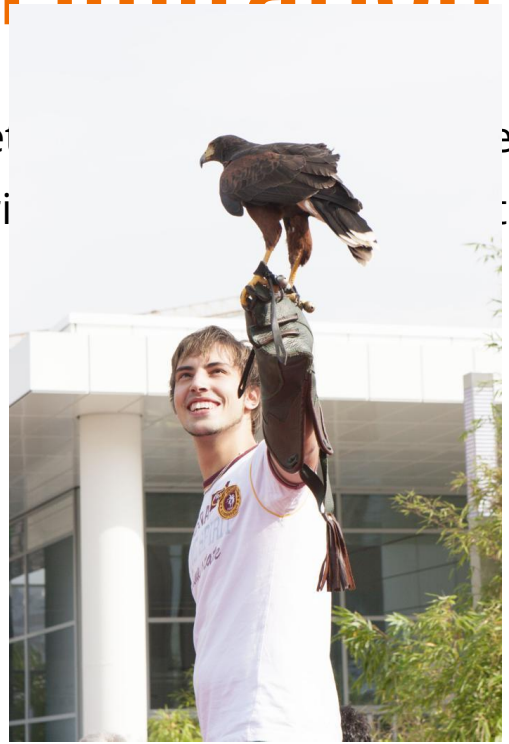
link to .csv

Limitation of Our Work

1. Our method tends to make some images overly white, resulting in a loss of their original saturation and contrast.

Limitation of Our Work

1. Our me e ima g in a loss of their ori rast.





Limitation of Our Work

1. Our method tends to make some images overly white, resulting in a loss of their original saturation and contrast.
2. A model that initially performed well on overexposed images showed improved results for underexposed images after adding the fuse mechanism. However, it caused a slight degradation in the performance for overexposed images (though still better than the original), making the model less flexible.



1. Our … a loss of their …
2. A mo… …ved impr… mechanism. However, it caused a slight degradation in the performance



Limitation of Our Work

1. Our method tends to make some images overly white, resulting in a loss of their original saturation and contrast.
2. A model that initially performed well on overexposed images showed improved results for underexposed images after adding the fuse mechanism. However, it caused a slight degradation in the performance for overexposed images (though still better than the original), making the model less flexible.
3. The fuse mechanism amplifies the artifacts inherent in Zero-DCE.

Limitation of Our Work

1. Our [illegible] a loss of
   thei[illegible]
2. A m[illegible] ved
   imp[illegible]
   med[illegible] nance
   for [illegible] ing the
   mod[illegible]
3. The [illegible]



Apply The Method To Practical Use

Exposure correction is highly useful for real-world applications.

Since it is common to have poorly exposed photos when taking pictures, our method can be used to correct the exposure of improperly exposed images, resulting in visually pleasing and well-balanced photos, as demonstrated in the results section.

Github link

Link: https://github.com/ChuEating1005/Exposure-Correction

Reference

- Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement
  github
- Learning Multi-Scale Photo Exposure Correction
  github
- Reversed and Fused Zero-DCE

Contribution

111550093 朱驛庭 25%

Develope Framwork, Integral Backward & Fuse LE Curve.

111550106 何義翔 25%

Integral Laplacian pyramid into Zero-DCE, Training & Testing.

111550040 曾紹幃 25%

Baseline implement, PPT design, Quantitative matrics implement.

111550086 陳冠智 25%

Dataset Sorting, ppt design, Idea of fuse, Quantitative and Quantitative analysis.

Thank You